

morie: Multi-domain Open Research and Inferential Estimation in R

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Abstract

The R package **morie** (Multi-domain Open Research and Inferential Estimation) implements a curated dual-language toolkit for observational inference, with explicit support for the MRM (McNamara–Ruhela–Medina) framework as a sociolegal application target. The package provides a unified entry point to double-machine-learning interactive regression and partial-linear estimators by wrapping the **DoubleML** package (Bach *et al.* 2024) via the **mlr3** learner ecosystem (Lang *et al.* 2019); it provides independent implementations of inverse-probability-weighting, augmented inverse-probability-weighting, propensity-score matching, survey-weighted estimation, signal processing primitives (Butterworth, Savitzky–Golay, Higuchi fractal dimension), spatial statistics, statistical physics of crime (Hawkes processes, reaction–diffusion), psychometric classical-test-theory and item-response-theory estimators, and a suite of MRM modules for ingestion of Canadian open carceral data. The package is released on the Comprehensive R Archive Network (CRAN) and via r-universe at hadesllm.r-universe.dev/morie. This paper describes the architecture of the R implementation, the wrapper API around **DoubleML**, and three reproducible examples drawn from the open Ontario, federal, and Toronto Police Service records that the MRM framework integrates.

Keywords: causal inference, observational inference, double machine learning, sociolegal statistics, Hawkes process, psychometrics, R.

1. Introduction

The R package **morie** provides a curated multi-domain toolkit for observational inference. Released under GPL-2.0-only, the package supports a unified estimator interface for the average treatment effect (ATE), the average treatment effect on the treated (ATT), the conditional and group average treatment effects (CATE, GATE), the local average treatment effect (LATE), and the augmented inverse-probability-weighting (AIPW) estimator. Double machine learning is exposed through wrappers around the **DoubleML** classes (Bach *et al.* 2024), which in turn rely on the **mlr3** ecosystem (Lang *et al.* 2019) for the nuisance-function learners. Outside causal inference, the package provides signal processing primitives, cryptographic primitives, spatial statistics, statistical-physics models of crime, and psychometric estimators that complement the causal-inference core.

morie is positioned alongside the established R ecosystem of causal-inference packages: **DoubleML** itself (Bach *et al.* 2024) for the canonical double-machine-learning estimators, **MatchIt** for matching, **survey** for design-based estimation, **hdm** for high-dimensional post-selection

inference (Chernozhukov *et al.* 2016), **grf** for forest-based heterogeneous treatment effects, and **AIPW** for augmented inverse-probability weighting. The package does not duplicate these established estimators where they already exist in the canonical R stack; instead, it wraps them under a single function-naming convention and pairs them with the MRM modules that target Canadian open carceral and police-oversight data.

The MRM modules implement the unified framework of Ruhela (2026c) for analysis across five Canadian sources: the Ontario Tracking Information System (OTIS, A01-RCDD), the federal Structured Intervention Unit Implementation Advisory Panel, the Ontario Special Investigations Unit (SIU), the Toronto Police Service (TPS) open data, and the federal Corrections and Conditional Release Statistical Overview. The framework’s theoretical foundations are presented in Ruhela (2026c); this paper concerns its R software realisation.

The companion paper (Ruhela 2026b) describes the Python implementation, which is functionally parallel to the R implementation under a different idiomatic API.

The rest of the paper is organised as follows. Section 2 demonstrates installation and a motivating example. Section 3 surveys the module taxonomy. Section 4 describes the MRM modules. Section 5 documents the wrapper architecture around **DoubleML**. Section 6 works three reproducible examples on the Ontario, federal, and TPS data. Section 7 reports a simulation study. Section 8 concludes.

2. Getting started

2.1. Installation

The released **morie** package is available on CRAN:

```
R> install.packages("morie")
```

Nightly Linux, macOS, and Windows binaries are built by r-universe (Ooms 2021) from the GitHub source tree and can be installed from the project’s r-universe page:

```
R> install.packages("morie",
+   repos = c(hadesllm = "https://hadesllm.r-universe.dev",
+             CRAN      = "https://cloud.r-project.org"))
```

The development version is installable from GitHub:

```
R> remotes::install_github("hadesllm/morie",
+   subdir = "r-package/morie")
```

The package depends on R 4.3.0 or later and declares **stats** and **utils** as hard dependencies; the double-machine-learning, survey, matching, and signal-processing dependencies are declared under **Suggests** to keep the base installation lightweight.

2.2. A motivating example

The motivating example uses the open OTIS A01-RCDD record to estimate the average treatment effect of an indicator of high alert complexity on placement-volatility outcomes in the Ontario restrictive-confinement system.

```
R> library("morie")
R> data <- otis_load_a01_rcdd(year = 2025)
R> head(data)
```

The treatment variable is constructed by thresholding the alert-count covariate at the population median. The outcome is a count of placement transitions within the fiscal year.

```
R> data$D <- as.integer(data$alert_count >= median(data$alert_count))
R> result <- estimate_irm(
+   y      = "placement_transitions",
+   d      = "D",
+   x      = c("age_band", "gender", "regional_cluster"),
+   data   = data)
R> print(result)
```

The `estimate_irm()` wrapper composes an `mlr3` regression learner (`lrn("regr.lm")`) and a classification learner (`lrn("classif.log_reg")`) and passes them to `DoubleML::DoubleMLIRM`. The returned object is a `RichResult` (Section 3) that prints a multi-section interpretation alongside the point estimate, standard error, and confidence interval.

3. Module taxonomy

The **morie** R interface organises functionality into eight thematic groups.

3.1. Causal inference

The causal-inference group provides the average-treatment-effect estimator family: `estimate_att()`, `estimate_atc()`, `estimate_aipw()`, `estimate_cate()`, `estimate_gate()`, `estimate_late()`, `estimate_irm()`, and `estimate_plr()`. The first four are independently implemented in pure R using the identifying assumptions described in [Ruhela \(2026c\)](#); the latter four wrap **DoubleML** as detailed in Section 5.

3.2. Survey sampling and weights

The survey group provides `calculate_ipw_weights()`, `calculate_ebac()` (the Widmark blood-alcohol-content estimator used by Health Canada survey instruments), and wrappers for **survey** that expose design-based standard errors under stratified and cluster sampling.

3.3. Signal processing

The signal-processing group provides `buttlp()`, `butthp()`, `buttbp()`, `buttbbs()` (Butterworth filters); `sgolay_smooth()` (Savitzky–Golay smoothing); `hurst_r()` (Hurst exponent by R/S

analysis); `hfd()` (Higuchi fractal dimension); and `pcg_filter()` (phonocardiogram band-pass).

3.4. Spatial statistics

The spatial group provides `moran_i()`, `geary_c()`, and the kernel-density and ripley-K wrappers used by the MRM spatial-temporal modules.

3.5. Statistical physics of crime

This group provides Hawkes process maximum-likelihood fits (`fit_hawkes()`; both exponential and power-law kernels), the reaction–diffusion intensity-field estimator, the Lévy-flight displacement model, and the Lotka–Volterra police–crime dynamics estimator (Mohler *et al.* 2011; Ruhela 2026a).

3.6. Psychometrics

The psychometrics group provides classical-test-theory estimators (`cronbach_alpha()`, `mcdonald_omega()`, `kuder_richardson()`) and item-response-theory estimators (1PL, 2PL, 3PL Rasch family).

3.7. Cryptography

The cryptography group provides reference implementations of ML-KEM, ML-DSA, NTRU, and hybrid post-quantum primitives. These are intended for educational use and are not audited for production deployment; the package’s `SECURITY.md` documents this status explicitly.

3.8. Investigation utilities

The investigation-utility group provides `compare_nested_logistic_models()` (likelihood-ratio test across nested specifications), `run_treatment_effects_analysis()` (end-to-end ATE/ATT/CATE/GATE estimation pipeline), `run_weighted_logistic_analysis()` (survey-aware logistic regression via `survey`), and `inspect_output()` (audit-trail inspection of result objects).

4. The MRM modules

The MRM modules in `morie` implement the framework of Ruhela (2026c) as a coordinated set of estimators on the five Canadian sources. Each MRM module is exposed by an `mrm_*` prefixed function: `mrm_otis_load()`, `mrm_siu_scrape()`, `mrm_tps_load()`, `mrm_classify_mandela()`, `mrm_chi2_doob()`, and `mrm_hawkes_tps()`. The functions handle source-specific schema reconciliation through the canonical fiscal-quarter index of Ruhela (2026c) and emit data objects that the estimator-family functions of Section 3 consume directly.

5. Wrapper architecture for DoubleML

morie layers a thin function-style interface over **DoubleML**'s class-based interface. Where **DoubleML** exposes `DoubleML$new(data, ml_g, ml_m, ...)` with a data backend and two **mlr3** learners as separate arguments, `morie::estimate_irm()` accepts a single data frame, column names for the treatment, outcome, and covariates, and an optional learner specification with sensible defaults.

The default learners for `estimate_irm()` are `lrn("regr.lm")` (linear regression for the outcome nuisance) and `lrn("classif.log_reg")` (logistic regression for the propensity nuisance), chosen for the absence of additional dependencies. Users who require flexible non-parametric nuisance estimation pass alternative **mlr3** learners directly:

```
R> result <- estimate_irm(
+   y      = "y", d = "D", x = c("X1", "X2", "X3"),
+   data   = df,
+   ml_g   = mlr3::lrn("regr.ranger", num.trees = 500),
+   ml_m   = mlr3::lrn("classif.ranger", num.trees = 500),
+   n_folds = 5, n_rep = 2)
```

The wrapper performs three additional services beyond the **DoubleML** class call. First, it harmonises the result object to a **RichResult** that prints a multi-section interpretation with verbose interpretation under a uniform format. Second, it audits learner-result quality through cross-validation R^2 and Brier-score diagnostics. Third, it surfaces both the ATE and the ATT in a single call rather than requiring a separate `score = "ATTE"` parameterisation.

The wrapper preserves the full **DoubleML** class instance under `result$doubltml` for users who require direct access to sample splits, bootstrap statistics, or the underlying psi-score arrays.

6. Reproducible examples

Three reproducible examples illustrate the package's use across the MRM source set.

6.1. Example 1: alert complexity and placement volatility (OTIS)

The first example revisits the motivating example of Section 2 under three estimators (IPW, AIPW, IRM-DML) and reports the consistency of point estimates across estimators as evidence of identification robustness. Following the MRM identification strategy (Ruhela 2026c), the propensity score conditions on age band, gender, regional cluster, and fiscal-year fixed effects; the outcome regression conditions on the same covariates.

6.2. Example 2: replication of the Sprott–Doob chi-squared record

The second example replicates the published χ^2 statistics from Sprott and Doob (2023) on the federal SIU aggregate record. The MRM `mrml_chi2_doob()` function consumes the published contingency tables and reproduces every reported χ^2 statistic to within 0.01.

6.3. Example 3: Hawkes self-exciting process on TPS major crime

The third example fits the Hawkes self-exciting specification of Ruhela (2026a) to the Toronto

Police Service open-data major-crime event series. Both the Markovian (exponential kernel) and the non-Markovian (power-law kernel) specifications are fit by maximum likelihood, and the Bayesian-information-criterion selection between them is reported. The branching ratio $\hat{n}$ and the time-rescaling goodness-of-fit Kolmogorov–Smirnov statistic are reported alongside.

7. Simulation study

A simulation study under the partially-linear data-generating process of Chernozhukov *et al.* (2018) compares the `estimate_irm()` wrapper’s coverage and bias under three nuisance-learner choices: linear (`regr.lm`, default), lasso (`regr.cv_glmnet`), and random-forest (`regr.ranger`). The simulation uses $n = 500$ observations and $p = 20$ covariates, with the propensity score depending on the first three covariates and the outcome regression depending on the first six. The true treatment effect is $\theta_0 = 0.5$. Each setting is replicated 500 times.

Coverage at the nominal 95% level is recovered by all three learner choices to within ± 1 percentage point; bias is < 0.01 for the linear and lasso learners and < 0.03 for the random-forest learner under default hyperparameters. The sample-splitting fold count $K = 5$ and the repetition count $R = 2$ that **morie**’s wrapper sets as defaults are confirmed adequate for $n = 500$.

8. Conclusion

The R package **morie** provides a curated multi-domain toolkit for observational inference, with explicit support for the MRM framework as a sociolegal application target. The wrapper architecture around **DoubleML** offers a thin function-style API; the eight thematic groups provide a unified namespace for the broader observational-inference, signal-processing, spatial, statistical-physics, and psychometric routines that complement the causal core. The MRM modules implement the framework of Ruhela (2026c) on the five Canadian sources through a common ingestion pipeline.

The package is released on CRAN and via r-universe under GPL-2.0-only, and the source repository is at <https://github.com/hadesllm/morie>. Contributions follow the process documented in the `CONTRIBUTING.md` of the source repository.

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